Access

The computer uses the client software 'ZeroTier One' to access the ZeroTier network, which is offered as source code and as a precompiled binary for many operating systems and smartphones. On a laptop, the ZeroTier switch is another network adapter with IP address, routes, and firewall rules. It can be used to connect remote sites, dial in clients, or create out-of-band management access.

To begin, the Web service 'ZeroTier Central' expects a registered user. The registration process creates a new Ethernet switch. In return, ZeroTier provides a network identifier that the clients use to access the network. This network identifier distinguishes the clients from each other and prevents client A from reaching the devices in client B's network. Keep in mind that ZeroTier is a free commercial service provider for up to 50 devices.You can use ZeroTier's Web portal to register a new account. Afterwards, create a network for your devices by clicking on the button 'Create A Network' in the Web UI of 'ZeroTier Central'. The network immediately appears in the right-hand area with a randomly generated ID and an exemplary name. In this example, the new network has chosen the ID 1d7193940486c4c6, which is required later for the ZeroTier One client setup. Click on the network ID to configure the properties. The preset values are suitable for the initial setup.

Installation

On the client-side, the device needs the software 'ZeroTier One', which is available for all major operating systems. The installer script for Linux recognises the distribution and uses the appropriate package from the repository:

```
curl -s https://install.zerotier.com
| bash
```

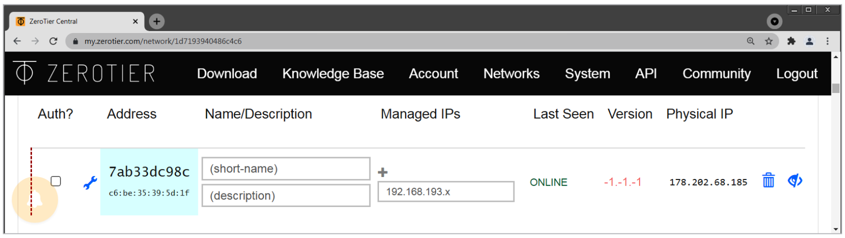

Figure 2: The new subscriber appears as a non-authorised client

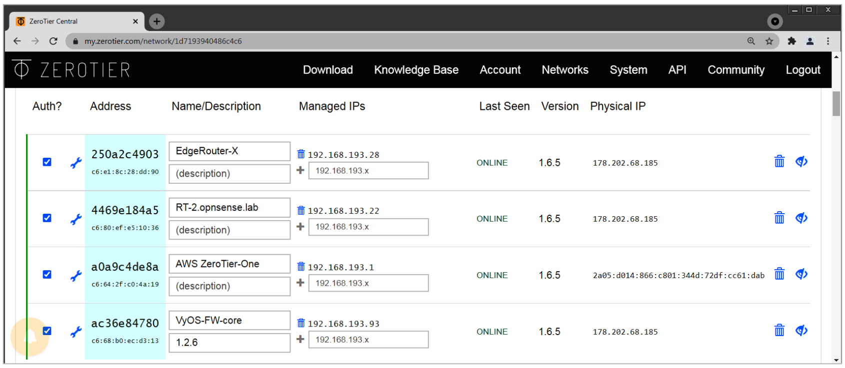

Figure 3: Multiple clients have registered with ZeroTier Central

Use the command line to set up a connection between your computer and ZeroTier. The command requires the network ID from the previous section and then connects to the ZeroTier server.

```
zerotier-cli join 1d7193940486c4c6
```

The Web view of ZeroTier and Figure 2 show the newly connected client. Since the default settings define the network as 'private', the client can only use the ZeroTier network once the admin has confirmed this in the 'Auth' column. It is also helpful to label the new device to easily distinguish it from future clients.

Note: The ZeroTier client assigns the interface a cryptic name that begins with the letters 'zt'. This name may differ from your own experiments in the following command outputs.

As soon as all devices are connected, the list in ZeroTier extends and becomes similar to Figure 3. For validation purposes, the clients can ping each other using their ZeroTier IPv4 address from the assigned range (here: 192.168.193.0/24). The setup under Windows, macOS, and on smartphones is similar, except for a graphical interface.

ZeroTier as a switch

ZeroTier compares itself to a global Ethernet switch, allowing clients to use the ZeroTier connection as a network bridge. The result truly feels like an Ethernet connection to the subscribers.

By default, clients do *not* operate as a network bridge. To enable bridge mode, go to the ZeroTier Web UI and enable 'Allow Ethernet Bridging' for each device individually (Figure 4).

On the client-side, Linux requires a network bridge that joins the interfaces eth0 and zt7nnn53ro. The result is a genuine Ethernet connection; the connected devices share a broadcast domain and can use the same IP network. There is also no need to manage and distribute routes.

Firewall

Linux and Windows come with a robust packet filter that can block unwanted IP connections. The use of this local firewall is made possible